

PCA Dimensionality Reduction: Numerical Walkthrough

Principal Component Analysis (PCA) simplifies high-dimensional data by projecting it onto a lower-dimensional subspace (e.g., compressing 2D down to 1D) while keeping the maximum amount of original spread (variance) of the shape.

The Sample Dataset

Imagine we have 3 points plotted on a 2D coordinate grid:

- **Point A:** (1, 2)
- **Point B:** (3, 4)
- **Point C:** (5, 6)

These points form a diagonal line. We want to compress them into a single **1D coordinate**.

Step-by-Step Mathematical Compression

Step 1: Mean Centering

We shift the data so its average center rests exactly at (0, 0).

1. Calculate averages: Mean X = 3, Mean Y = 4.
2. Subtract the means:
 - **Point A'** = (-2, -2)
 - **Point B'** = (0, 0)
 - **Point C'** = (2, 2)

Step 2: Compute the Covariance Matrix

Using the sample covariance formula (dividing by $N-1 = 2$):

1. $\text{Cov}(X, X) = \frac{(-2)^2 + 0^2 + 2^2}{2} = 4$
2. $\text{Cov}(Y, Y) = \frac{(-2)^2 + 0^2 + 2^2}{2} = 4$
3. $\text{Cov}(X, Y) = \frac{(-2 \times -2) + (0 \times 0) + (2 \times 2)}{2} = 4$

Our Covariance Matrix (\$S\$) is: $S = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$

Step 3: Find Eigenvalues & Eigenvectors

1. Solve $\det(S - \lambda I) = 0$: This yields: $\lambda_1 = 8$ (PC1, captures 100% of variance) and $\lambda_2 = 0$ (PC2).
2. Solve for the eigenvector $\mathbf{v}$ for $\lambda_1 = 8$: Normalizing this vector gives:
 $\mathbf{v} = \begin{pmatrix} 0.7071 \\ 0.7071 \end{pmatrix}$

Step 4: Projection to 1D

We calculate the dot product of centered coordinates with $\mathbf{v}$: $\text{Projected Coordinate} = (X_{\text{centered}} \times 0.7071) + (Y_{\text{centered}} \times 0.7071)$

- Point A' (-2, -2): $(-2 \times 0.7071) + (-2 \times 0.7071) = -2.83$
- Point B' (0, 0): $(0 \times 0.7071) + (0 \times 0.7071) = 0.00$
- Point C' (2, 2): $(2 \times 0.7071) + (2 \times 0.7071) = 2.83$

Original 2D Point	Centered 2D Point	Compressed 1D Coordinate	Information Retained
Point A (1, 2)	(-2, -2)	-2.83	100% (No loss)
Point B (3, 4)	(0, 0)	0.00	100% (No loss)
Point C (5, 6)	(2, 2)	2.83	100% (No loss)